Travel Planning Agent - High Level Design
User Input Layer
User queries, preferences, and context
Session Management
UserContext initialization & preferences
Input Guardrails Layer
Budget Analysis & Validation
Guardrail Feedback
Show warning & suggestions
Valid?
Yes
Main Travel Agent
Intent Classification & Task Routing
Tools: Weather Forecast | Handoffs: Specialized Agents
Flight Specialist
• Search flights
• Apply preferences
• Price optimization
→ FlightRecommendation
Hotel Specialist
• Search hotels
• Amenity matching
• Location analysis
→ HotelRecommendation
Conversational Specialist
• General queries
• Trip planning advice
→ Raw String Response
Weather Tool
• City forecasts
• Date-specific data
• Travel planning info
→ TravelPlan
Travel Planning Output
• Activity suggestions
• Budget analysis
→ TravelPlan
User Context
• User ID
• Preferred airlines
• Hotel amenities
• Budget level
• Session data
Output Formatting & Response Layer
Structured output processing based on response type
FlightRecommendation | HotelRecommendation | TravelPlan | Generic Response
Legend
Main Agent
Specialist Agent
Tools/Utilities
Guardrails
Context Flow
Formatted User Response
Personalized recommendations & insights